Documentation

Functional Description
Electronic governor
ECU 8
for MTU Series 1600
Genset application

E532291/01E

Power. Passion. Partnership.

Printed in Germany

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

ECU 8

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1 General

1.1 General requirements

1.1.1 General

In addition to the notices in this brochure, the valid, country-specific, statutory regulations on accident prevention and environmental protection are to be observed. This engine is built according to the current state of the art standards and valid regulations and rules. Nevertheless dangers to persons and property are still present if this engine is:

- • Not used properly
- • Operated, maintained and repaired by non-trained and unauthorized personnel
- • Modified or changed
- • Safety notices are not observed

1.1.2 Intended use

This engine is only to be used for the purpose stated in the contract or at the delivery. A different or additional use will be considered a “non-intended use”. The manufacturer assumes no responsibility for any damages resulting from a “non-intended use”. The user is solely responsible.

Intended use also includes observing the operating instructions and the maintenance and repairs specifications.

1.1.3 Modifications or changes

Any unauthorized modifications or changes to the engine affect safety.

MTU assumes no responsibility and accepts no warranty claims for damages resulting from unauthorized modifications or changes.

1.1.4 Spare parts

Only MTU original parts may be used when exchanging or replacing components or assemblies.
Liability and warranty claims against the engine manufacturer for damage resulting from the use of non-OEM parts will be rejected.

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1.2 Safety regulations for maintenance and repair work

1.2.1 Safety regulations for commissioning

The product must be properly installed and have passed the acceptance test according to the MTU regulations before it can be commissioned.

Each time the device or system is put into commission make sure

- • That all maintenance and repair work are finished
- • That all loose parts have been removed from rotating parts
- • That no one is in the danger zone of moving parts

Right after placing the device or system into commission make sure that the controls and indicators and the monitoring, signaling and alarm systems are operating properly.

Use of the overspeed input

For applications requiring functional safety (engine power before engine protection) the “overspeed” input is available. In addition, this input enables an engine start without regarding certain engine safety checks.

The overspeed function is recommended for plants with requirements for functional safety. Using this input above this may cause the risk of harm or destruction, beneath operation out of emission legislation. In addition, the are another risks, such as loose parts flying around based on the destruction by overspeed.

1.2.2 Safety regulations for operation

Regularly train and practice the operating procedures in case of emergencies.

The operator must be familiar with the controls and indicators.

The operator must know the effects of each operating step performed.

The operator must carry out each operating step according to the documentation.

During operation the display instruments and monitoring groups must be constantly monitored with respect to the current operating states, the operating limits and any warnings and alarms.

If a problem with the system is detected or if the system reports a problem:

- • The person(s) responsible is/are to be informed
- • The messages is to be evaluated
- • Any emergency actions are to be taken, as necessary, e.g. emergency engine stop

1.2.3 Engine operation

Always wear hearing protection when engine is running!

Make sure engine room is well ventilated!

Immediately wipe up any operating fluids that have leaked out or have been spilled; use a binding agent as necessary.

A combustion engine's exhaust is poisonous. Inhaling the exhaust is hazardous to your health.
Exhaust pipes must be air tight and routed to the open air.

Do not touch or come into contact with any battery terminals, generator terminals or cables while engine is running!

Improper insulation or grounding of electrical parts can lead to serious electrical shocks causing injury.

Absolutely do not loosen or open any water, oil, brake fluid, fuel, compressed air or hydraulic lines while the engine is running!

1.2.4 Maintenance and repairs

A major factor in safety is the compliance with the maintenance and repairs specifications.

Do not perform any maintenance or repair work while the engine is running, unless it is explicitly permitted. Secure the engine from being accidentally started. If an electrical starter is installed, disconnect the battery. If a

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compressed air starter is installed, close the main valve of the compressed air system and release pressure in the air line. Place a sign “Do not operate” in the operating room or at the controls! Keep all nonessential persons away!

Do not rectify any problems for which you do not have the proper knowledge or the special tools required! Maintenance and repair work should only be made by authorized and qualified personnel.

Only use appropriate and calibrated tools.

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Do not work on engines or components that are only being held up by lifting equipment or by a crane. Always properly support these components with suitable equipment before starting any maintenance or repair work.

Before cranking engine make sure no one is in the danger zone of the engine. After finishing work on the engine, make sure all protective devices are reinstalled and that all tools and loose parts have been removed from the engine.

Fluids exiting at high pressures can penetrate through clothing and skin and cause serious injuries. Before starting any work, release the pressure in those systems that are to be worked on!

Never rebend any fuel injection lines and do not install any lines that have been rebent. Keep fuel injection lines and connections clean. When removing or opening lines, always close off the openings using suitable caps and plugs.

Make sure to not damage the fuel lines during maintenance and repair work. When installing lines always use the correct tightening torque and make sure that all brackets and dampers are installed properly.

Makes sure that all fuel injection lines and oil pressure lines have sufficient clearance so as to avoid any contact with other components. Do not install fuel or oil lines near hot components unless the design does not allow otherwise.

Elastomers (e.g. Viton sealing rings) are safe at normal operating temperatures. If a fire occurs or if the temperature rises above 300 °C, the materials will start to melt setting off hydrogen fluoride fumes, which is an acid. If this acid comes into contact with the skin, serious burns can occur. If elastomer seals appear burnt or glassy, do not tough them with unprotected hands! Wear protective gloves!

Be cautious with hot fluids in lines, pipes and spaces ⇒ Risk of burn injuries!

Observe the cool off times of components that are heated up for installation/removal Risk of burn injuries!

Do not touch or come into contact with the hot parts of the compressor and the exhaust system ⇒ Risk of burn injuries!

Be careful when removing vent screws or screw plugs from the engine. To prevent the fluid from escaping, hold a rag over the vent screw or stop plug. The risk of injury is greater if the engine was just turn off and the fluids are still hot.

Be careful when draining hot operating fluids. ⇒ Risk of scalding!

Drain the operating fluids into a suitable container and clean up any spilled fluids using a binding agent as necessary.

Make sure the engine room is well ventilated when changing any operating fluids or working on the fuel system!

For any work above body height use safety gear and work platforms.
Make sure engine parts are placed on stable surfaces!

To avoid back injuries when lifting components, adults should only lift between 10 to 30 kg depending on age and gender, therefore:

- Get help or use a lifting aid.
- Make sure that all chains, hooks, loops, etc. are tested and approved and have sufficient payload rating and that the hooks are correctly positioned. Lifting lugs may not be loaded laterally.

When carrying out maintenance and repair work on the engine, pay special attention to cleanliness. After finishing the maintenance and repair work, make sure that any loose parts have been removed.

1.2.5 Welding

Welding is prohibited on the engine or on installed assemblies!

Never use the engine as a ground connection for welding! This will prevent the welding current from being conducted through the engine causing burns to the mounts, bearings and gears, which in turn leads to bearing seizures or damage to the components.

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